

GRADE 12 IT

EUEE MASTER STUDY GUIDE

Ethiopian University Entrance Exam
Complete Exam Preparation

Information Systems · Emerging Tech · Databases · Web · Maintenance

Simplified · Exam-Focused · High-Score Strategies

UNIT 1

INFORMATION SYSTEMS AND THEIR APPLICATIONS

1.1 Ethical, Legal, Social, Environmental & Health Issues

Ethical issues:

- **Privacy:** right to control personal information; cookies, data harvesting
- **Accuracy:** responsibility to ensure information is correct
- **Property:** who owns digital content — creator, employer, or public?
- **Accessibility:** ensuring equal access to information technology

Legal issues:

- **Cybercrime:** hacking, phishing, identity theft, online fraud
- **Copyright law:** protects original creative works from unauthorised copying
- **Data protection:** laws governing how personal data must be stored and used
- **Digital signature:** legally binding electronic signature

Social issues:

- **Digital divide:** gap between those with/without access to technology
- **Online addiction:** excessive internet/social media use affecting mental health
- **Cyberbullying:** harassment using digital means
- **Social isolation:** reduced face-to-face interaction

Environmental issues:

- **E-waste:** discarded electronics containing toxic materials (lead, mercury)
- **Energy consumption:** data centres use massive electricity; carbon footprint
- **Green computing:** designing energy-efficient systems

Health issues:

- **RSI (Repetitive Strain Injury):** wrist/arm pain from repetitive keyboard/mouse use
- **Eye strain (CVS):** computer vision syndrome from screen glare
- **Ergonomics:** designing workspaces to reduce physical strain

1.2 Intellectual Property

Definition — Intellectual Property (IP)

Legal rights over creations of the mind: inventions, literary/artistic works, symbols, names, and designs.

- **Copyright:** automatic protection for original works (books, music, software)
- **Patent:** exclusive right to an invention for 20 years
- **Trademark:** protects brand names, logos, symbols
- **Trade secret:** confidential business information (Coca-Cola formula)
- **Open source:** software with freely available source code (Linux, Firefox)

1.3 Digital Identity Management

Definition — Digital Identity

The online representation of a person or entity. Includes usernames, passwords, email addresses, and digital certificates.

- **Authentication:** verifying who you are (passwords, biometrics, 2FA)
- **Authorisation:** what you are allowed to do after authentication
- **Multi-factor authentication (MFA):** combines something you know + have + are
- **Identity theft:** stealing someone's personal information to commit fraud

EXAM FOCUS

- Digital divide, cybercrime, copyright, and e-waste are common EUEE IT topics.
- Know the difference between copyright, patent, trademark — with examples.
- Ergonomics and health issues of IT use are testable.

UNIT 2

EMERGING TECHNOLOGIES

2.1 Big Data

Definition — Big Data

Extremely large datasets that traditional data processing tools cannot handle. Characterised by the 5 Vs.

Volume	Massive amount of data	Facebook generates 4 petabytes/day
Velocity	Speed of data generation	Real-time stock market data
Variety	Different formats	Text, video, audio, images, sensor data
Veracity	Data accuracy/trustworthiness	Removing noise and errors
Value	Usefulness of data	Insights driving business decisions

Applications of Big Data:

- Healthcare: predicting disease outbreaks, personalised medicine
- Banking: fraud detection, credit scoring
- Retail: recommendation systems (Amazon, Netflix)
- Government: smart cities, traffic management

2.2 Cloud Computing

Definition — Cloud Computing

Delivering computing services (servers, storage, databases, networking, software) over the internet ("the cloud") on a pay-per-use basis.

Three service models:

- **IaaS (Infrastructure as a Service):** virtual machines, storage, networking (AWS, Microsoft Azure)
- **PaaS (Platform as a Service):** development environment (Google App Engine, Heroku)
- **SaaS (Software as a Service):** software accessed via browser (Gmail, Dropbox, Salesforce)

Four deployment models:

- **Public cloud:** shared infrastructure, managed by provider (AWS, Google Cloud)

- **Private cloud:** dedicated to one organisation; more secure
- **Community cloud:** shared by organisations with common needs
- **Hybrid cloud:** mix of public and private cloud

Benefits: cost savings, scalability, accessibility, automatic updates, disaster recovery

Limitations: internet dependency, security/privacy concerns, vendor lock-in, downtime risk

2.4 Internet of Things (IoT)

Definition — IoT (Internet of Things)

A network of physical objects ("things") embedded with sensors, software, and connectivity to exchange data with other devices and systems over the internet.

Examples:

- Smart home devices (thermostats, security cameras, smart TVs)
- Wearables (smartwatches, fitness trackers)
- Industrial IoT (factory automation, supply chain tracking)
- Smart agriculture (soil sensors, automated irrigation)
- Smart cities (traffic lights, waste management, energy grids)

How IoT works:

- Sensors collect data → transmit via internet → cloud processing → action or insight

Challenges: security vulnerabilities, data privacy, interoperability, bandwidth

EXAM FOCUS

- IaaS vs PaaS vs SaaS — know what each provides and real examples.
- Big Data 5 Vs — Volume, Velocity, Variety, Veracity, Value.
- IoT definition and practical applications are EUEE favourites.

UNIT 3

DATABASE MANAGEMENT SYSTEM (DBMS)

3.1 Relational Database Concepts

Definition — Database

An organised collection of structured data stored and accessed electronically. A DBMS is software that manages databases (e.g. MySQL, Oracle, Microsoft Access).

Key terms:

- **Table (Relation):** data organised in rows and columns
- **Record (Row/Tuple):** a single entry in a table
- **Field (Column/Attribute):** a category of data
- **Primary Key:** unique identifier for each record (no duplicates, no nulls)
- **Foreign Key:** field linking one table to another
- **Query:** request to retrieve specific data

Advantages of DBMS over file system:

- Reduces data redundancy, ensures data integrity, easy data sharing
- Better security, centralised control, data independence

3.2 SQL — Structured Query Language

SQL is divided into three sub-languages:

- **DDL (Data Definition Language):** creates and modifies database structure
- **DML (Data Manipulation Language):** manipulates data within tables
- **DQL (Data Query Language):** retrieves data (SELECT)

Essential SQL commands:

CREATE TABLE	Create a new table	CREATE TABLE Students (ID INT PRIMARY KEY, Name VARCHAR(50));
INSERT INTO	Add a new record	INSERT INTO Students VALUES (1, 'Abebe');
SELECT	Retrieve data	SELECT * FROM Students WHERE Name='Abebe';

UPDATE	Modify existing data	UPDATE Students SET Name='Kebede' WHERE ID=1;
DELETE	Remove a record	DELETE FROM Students WHERE ID=1;
DROP TABLE	Delete entire table	DROP TABLE Students;
ALTER TABLE	Modify table structure	ALTER TABLE Students ADD Age INT;

SELECT query with clauses:

**SELECT column1, column2 FROM table WHERE condition ORDER BY column
GROUP BY column HAVING condition;**

EXAM FOCUS

- SQL commands — CREATE, INSERT, SELECT, UPDATE, DELETE — all likely tested.
- Primary key: unique + not null. Foreign key: links tables.
- WHERE filters rows; HAVING filters groups; ORDER BY sorts.

PRACTICE QUESTIONS

1. What is a primary key? How does it differ from a foreign key?
2. Write SQL to: (a) Create a table called Employees (b) Insert a record (c) Select all records where salary > 5000
3. Distinguish between DDL and DML with examples.
4. What are the advantages of a DBMS over a traditional file system?

UNIT 4

WEB AUTHORING

4.1 XML — eXtensible Markup Language

Definition — XML

A markup language that defines rules for encoding documents in a format readable by both humans and machines. Used to transport and store data.

XML structure:

[declaration]

data [nested elements]

XML rules:

- Must have one root element
- All tags must be closed
- Tags are case-sensitive
- Attributes must be quoted
- Elements must be properly nested

XML vs HTML:

Purpose	Store and transport data	Display data in browser
Tags	User-defined	Predefined (h1, p, div, etc.)
Case sensitivity	Case-sensitive	Not case-sensitive
Closing tags	Always required	Sometimes optional
Focus	Data structure	Presentation

4.3 Publishing a Website

Steps to publish a website:

1. Register a domain name (e.g. mysite.com)
 2. Choose a web hosting service
 3. Upload files via FTP (File Transfer Protocol)
 4. Test the website in multiple browsers
- **URL structure:** http://www.sitename.com/page.html
 - protocol:// domain name / path

UNIT 5

MAINTENANCE AND TROUBLESHOOTING

5.1-5.2 Software Installation and Troubleshooting

Software installation steps:

1. Check system requirements (OS, RAM, disk space, CPU)
2. Download from official source or use physical media
3. Run installer, accept license, choose directory
4. Complete installation and restart if required

Common software problems and solutions:

Software crashes	Corrupted files, insufficient RAM	Reinstall, add RAM
Slow performance	Many background apps, full disk	Close apps, disk cleanup, defrag
Won't install	Insufficient space/rights	Free disk space, run as admin
Compatibility error	Wrong OS version	Check system requirements

5.3 Tools to Speed Up Computer

- **Disk Defragmenter:** reorganises fragmented files on HDD; not needed for SSD
- **Disk Cleanup:** removes temporary files, recycle bin contents, unused files
- **Windows Update:** patches security vulnerabilities, improves stability
- **System Restore:** returns system to a previous working state using restore points
- **Task Manager:** monitor running processes, end unresponsive apps, check performance

5.4 Network Troubleshooting

Common network problems:

- No internet connectivity, slow speeds, intermittent connection, DNS errors

Troubleshooting steps:

1. Check physical connections (cables, router/modem lights)

2. Restart modem/router
3. Check IP configuration (ipconfig / ifconfig)
4. Ping test (ping 8.8.8.8 to test internet; ping gateway to test LAN)
5. Check DNS settings
6. Check firewall settings

Key command-line tools:

- **ipconfig/ifconfig:** shows IP address, subnet mask, default gateway
- **ping:** tests connectivity to a host (measures round-trip time)
- **tracert/traceroute:** shows path packets take to a destination
- **nslookup:** queries DNS to find IP address of a domain name
- **netstat:** shows active network connections

EXAM FOCUS

- SQL SELECT with WHERE, ORDER BY — write full queries in exams.
- Cloud service models: IaaS/PaaS/SaaS — know what each provides.
- Network troubleshooting commands: ping, ipconfig, tracert, nslookup.

LAST-MINUTE REVISION GUIDE

Digital divide	Gap between those with and without access to technology
Copyright	Automatic legal protection of original creative works
Patent	Exclusive right to an invention for 20 years
Big Data 5 Vs	Volume, Velocity, Variety, Veracity, Value
IaaS	Virtual infrastructure (AWS, Azure)
PaaS	Development platform (Google App Engine)
SaaS	Software via browser (Gmail, Dropbox)
IoT	Physical objects connected to internet sharing data
Primary key	Unique identifier; no nulls; no duplicates
Foreign key	Links one table to another
SELECT syntax	SELECT cols FROM table WHERE cond ORDER BY col
DDL examples	CREATE, ALTER, DROP

DML examples	INSERT, UPDATE, DELETE
XML vs HTML	XML: stores data (custom tags); HTML: displays data (fixed tags)
Ping command	Tests network connectivity; measures round-trip time
Disk Defrag	Reorganises HDD files; improves read speed

EUEE EXAM STRATEGY

Time Management

- **Skim first (5 min):** Read all questions. Mark easy (E), medium (M), hard (H).
- **Easy first:** Answer all E questions to bank safe marks and build confidence.
- **Pace yourself:** Aim for ~2 minutes per question. If stuck over 3 minutes, skip and return.
- **Show working:** Even wrong answers earn partial marks if steps are correct.
- **Final review:** Use last 10 minutes to check calculations, names, and dates.

Answering Techniques

- Multiple-choice: eliminate obviously wrong answers first, then choose from what remains.
- Definition questions: always include the key term in your answer.
- Explanation questions: structure as Definition → Example → Importance.
- Diagram questions: label all parts, use arrows correctly, add units where needed.
- Essay questions: write a clear intro, 3-4 main points with evidence, strong conclusion.

Common Exam Traps

- Confusing similar terms — always define the exact term asked.
- Incomplete answers — re-read every question before finalising.
- Time overrun — if a question takes too long, note it and move on.
- Skipping diagrams — a well-labelled diagram often earns more marks than a paragraph.
- Not reading all options in MCQ — always read all four choices before selecting.